

AnchorIQ

Customer Churn & Retention Analysis

Dataset: telco_churn.csv

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7,043 Records	21 Features	Churn Target	0 Fixed Nulls	0 Dupes Cleared
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1. EXECUTIVE SUMMARY & AI INSIGHTS

Executive Summary

Your customer base of 7,043 shows a 26.5 % churn rate (1,869 “Yes” cases). The data reveal clear risk groups: new customers (tenure 0-9), those on month-to-month contracts, fiber-optic subscribers, and high-charge users. A logistic regression model (AUC 0.8414) can flag the 10 % most-at-risk customers, catching 29 % of all churners while touching only 704 people. Acting on these insights can reduce churn and save money.

Key Findings

- Overall churn** – 26.5 % (1,869/7,043).
- Tenure 0-9 months** – 49.8 % churn (1.9× the overall rate) across 1,854 customers (26 % of the dataset).
- Fiber-optic service** – 41.9 % churn (1.6× the overall rate) among 3,096 customers (44 % of the dataset).
- Month-to-month contract** – 42.7 % churn (1.6× the overall rate) among 3,875 customers (55 % of the dataset).
- Top 10 % risk predictions** – 704 customers capture 29 % of all churners, a 2.9× improvement over random selection.
- Model performance** – Logistic regression gives 73 % recall and 56 % precision on unseen data (AUC 0.8414).

Recommendations

- Target the top 704 high-risk customers**

Why? They contain 29 % of all churners.

Impact? A focused retention offer could reach nearly a third of churn cases while only contacting 10 % of the customer base.

Assumption: The offer will reduce churn in this group by a measurable amount.

- Focus on month-to-month contract holders**

Why? 42.7 % churn rate, 55 % of customers.

Impact? A 10 % relative drop would save about 166 churn cases.

Assumption: A contract upgrade or loyalty incentive will lower the rate.

- Address fiber-optic subscribers**

Why? 41.9 % churn rate, 44 % of customers.

Impact? A 10 % relative drop would prevent roughly 130 churn cases.

Assumption: Service quality or

2. DATASET OVERVIEW & QUALITY AUDIT

Audit Metric	Value	Status
Total Records	7,043	Validated
Total Feature Count	21	Validated
Numerical Columns	3	Active
Categorical Columns	18	Active
Target Column	Churn	Identified
Missing Cells Imputed	0	Resolved
Duplicate Rows Purged	0	Resolved
Data Quality Index	100.0%	Optimal

Preprocessing Pipeline Execution:

- Converted 'TotalCharges' to numeric
- Filled 11 nulls in 'TotalCharges' with median (1397.47)

3. MACHINE LEARNING BENCHMARKS

Problem Formulation: **Classification** | Best Champion: **Logistic Regression** | Training: **5,634** samples | Evaluation: **1,409** samples | Engineered Features: **19**

Model Candidate	AUC Score	Validation Status
Champion: Logistic Regression	AUC = 0.8414	Top Performer
Logistic Regression	AUC = 0.8449	Linear
Random Forest	AUC = 0.8423	Ensemble
XGBoost Classifier	AUC = 0.8408	Gradient Boosted
Accuracy	73.74%	Overall Match
Cross-Validation	0.8449 ± 0.006	5-Fold CV

Key Drivers & Influential Signals:

Rank	Feature Variable	Importance Weight
1	tenure	0.4783
2	InternetService	0.1578
3	Contract	0.1155
4	MonthlyCharges	0.1076
5	TotalCharges	0.0386
6	StreamingMovies	0.0181
7	StreamingTV	0.0171
8	OnlineSecurity	0.0140

4. EXPLORATORY DATA ANALYSIS (EDA CHARTS)

Figure 1: Feature Distributions

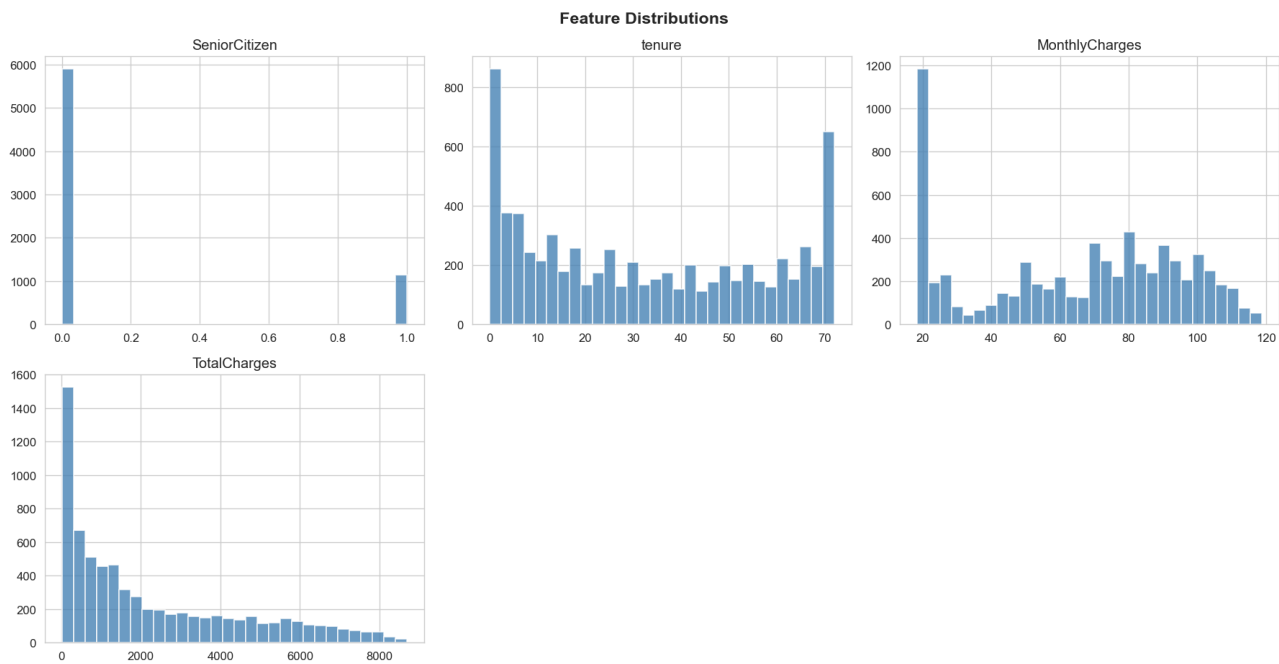


Figure 2: Feature Correlation Heatmap

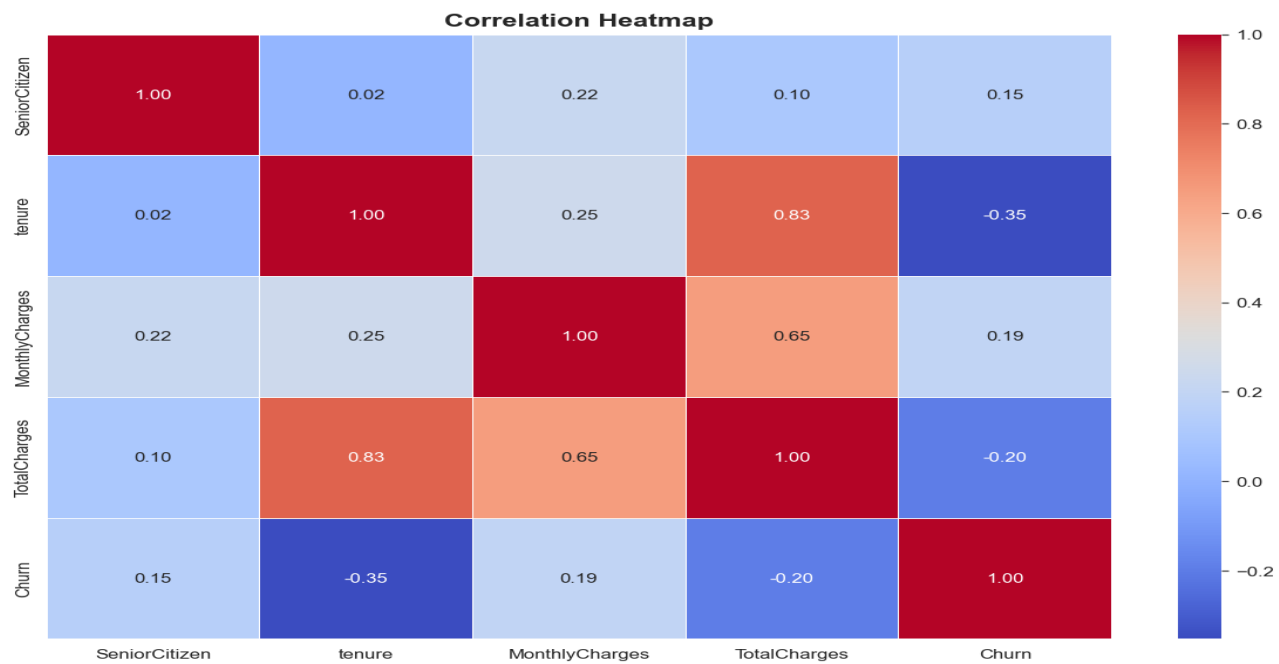


Figure 3: Target Variable Distribution

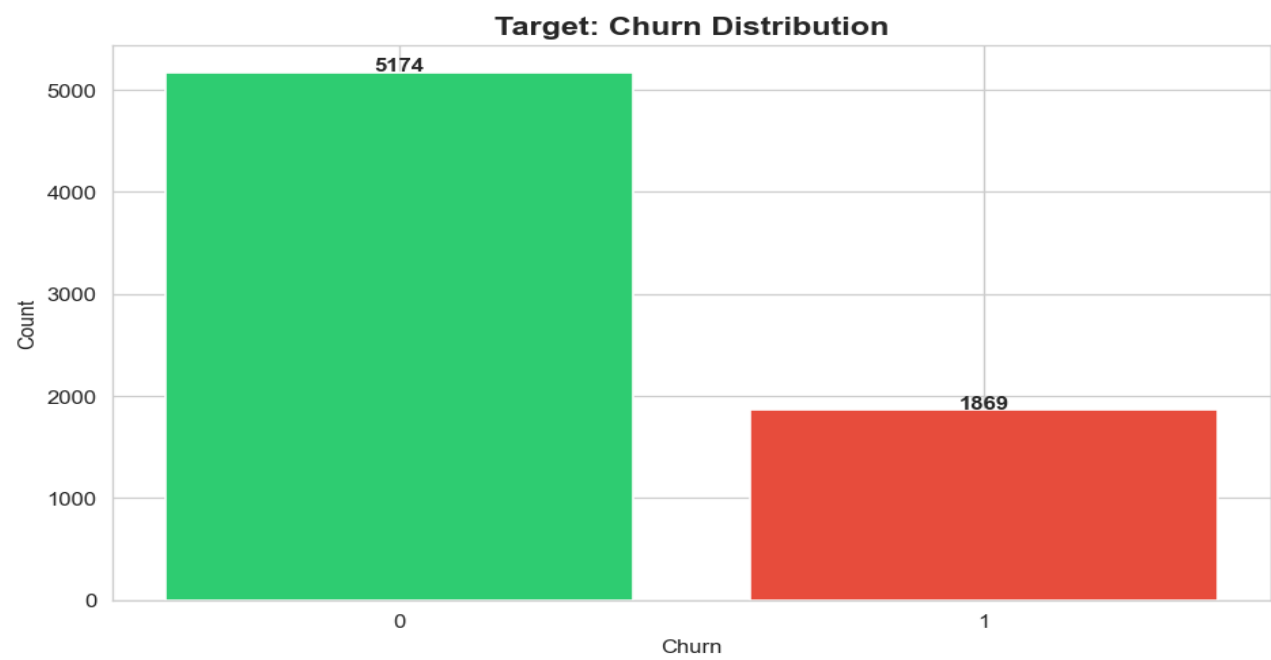


Figure 4: Categorical Feature Impact

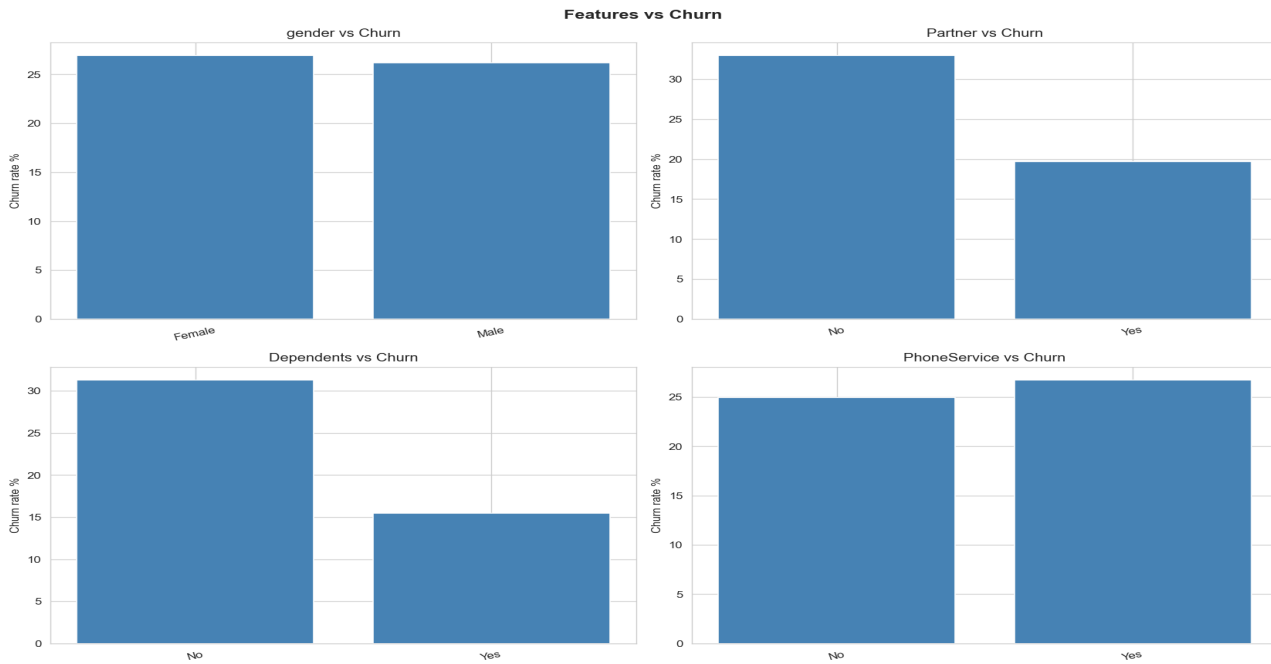
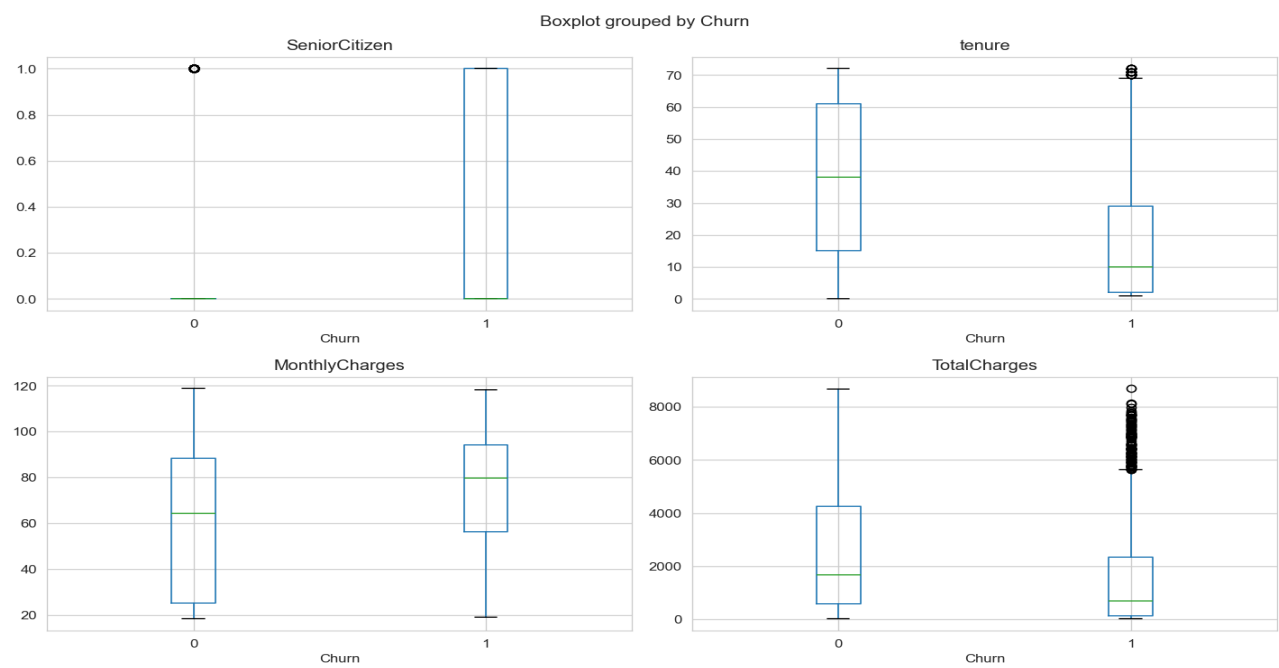


Figure 5: Numerical Distributions by Target



5. FEATURE STATISTICAL SUMMARY

Feature	Mean	Median	Std Dev	Min	Max	Skewness
SeniorCitizen	0.16	0.0	0.37	0.0	1.0	1.83
tenure	32.37	29.0	24.56	0.0	72.0	0.24
MonthlyCharges	64.76	70.35	30.09	18.25	118.75	-0.22